Q1. Name some popular free and open-source database management systems and state their type (relational, NoSQL, etc.). (BL3, LO4)

Database Management Systems (DBMS) are software applications that help store, organize, and manage data efficiently. Open-source DBMS solutions are widely used in organizations because they are cost-effective, community-supported, and customizable.

Database System	Type	Description
MySQL	Relational (RDBMS)	A widely used open-source SQL database for web and enterprise applications.
PostgreSQL	Relational (RDBMS)	Advanced database supporting SQL and JSON queries; known for reliability.
SQLite	Relational (RDBMS)	Lightweight, file-based database used in embedded systems and mobile apps.
MongoDB	NoSQL (Document-based)	Stores data in flexible JSON-like documents for scalability and speed.
Cassandra	NoSQL (Column-based)	Designed for distributed and high-volume data storage.
Redis	NoSQL (Key-value store)	In-memory data store often used for caching and real-time analytics.

Q2. List the steps to set up monitoring of server log files using the Nagios Open-Source Network Monitoring tool. (BL3, LO5)

Nagios is a powerful open-source monitoring tool used to track the status of network devices, servers, and applications. Setting up Nagios to monitor server log files involves several steps:

1. Install Nagios Core:
   Download and install Nagios Core on a Linux system (e.g., Ubuntu or CentOS). This provides the central monitoring framework.

2. Install NRPE (Nagios Remote Plugin Executor):
   NRPE allows Nagios to execute monitoring plugins on remote servers, enabling log monitoring on multiple hosts.

3. Install Necessary Plugins:
   Add the check_logfiles plugin, which helps scan log files for patterns such as "ERROR" or "WARNING".

4. Configure Command Definitions:
   Define monitoring commands in the commands.cfg file, specifying what logs or keywords Nagios should check for.

5. Create Host and Service Definitions:
   In the configuration files (hosts.cfg and services.cfg), define the servers (hosts) and specific log monitoring services.

6. Specify Log File Paths:
   Indicate the log files to monitor (e.g., /var/log/syslog) and the patterns to search for in the plugin configuration.

7. Restart Nagios Service:
   Apply the new configurations by restarting Nagios with:

8. sudo systemctl restart nagios

9. View and Analyze Alerts:
   Use the Nagios web interface to visualize alerts and view real-time log status updates.

Q3. Choose the best open-source monitoring and alerting tool for cloud-hosted applications and briefly explain why, using Prometheus as an example. (BL3, LO5)

Best Tool: Prometheus

Prometheus is the most widely used open-source monitoring and alerting tool for cloud-native applications.

Reasons for Choosing Prometheus:

- **Time-Series Data Model**:
  Prometheus stores all metrics as time-series data, allowing for efficient performance tracking and historical trend analysis.
- **Pull-Based Data Collection**:
  Unlike other tools, Prometheus pulls metrics from endpoints via HTTP, making it highly efficient for dynamic cloud environments like Kubernetes.
- **Powerful Query Language (PromQL):**
  PromQL allows users to query, aggregate, and analyze metrics in real time, supporting complex operational insights.
- **Integration with Grafana**:
  Prometheus integrates seamlessly with Grafana, a visualization tool used for creating rich dashboards and charts.
- **Alertmanager for Notifications**:
  Prometheus includes an Alertmanager component that sends alerts via email, Slack, or webhook when thresholds are breached.
- **Cloud-Native and Scalable**:
  It is ideal for microservices and Kubernetes environments due to built-in service discovery and minimal configuration.

Q4. How can contributors participate in the Mozilla Firefox open-source project, and what are the basic steps involved? (BL3, LO6)

Mozilla Firefox is an open-source web browser maintained by the global Mozilla community. Anyone can contribute to its development, documentation, testing, or localization.

Ways to Contribute:

- Writing or reviewing code.
- Testing new builds and reporting bugs.
- Translating Firefox into different languages.
- Enhancing documentation and tutorials.

Basic Steps to Contribute:

1. **Create a Mozilla Account:**
   Register on Mozilla's platforms (Bugzilla and Phabricator) to track issues and submit contributions.
2. **Set Up the Development Environment:**
   Follow Mozilla's build setup guide to install dependencies and clone the Firefox source code using:
   hg clone https://hg.mozilla.org/mozilla-central
3. **Find a "Good First Bug":**
   Search for beginner-friendly issues labelled "Good First Bug" on Bugzilla.
4. **Make Code Changes:**
   Modify the source code as per the bug report and test it locally using the build tools.

5. **Submit a Patch:**
   Upload the patch for review through Phabricator or Bugzilla.
6. **Participate in Code Reviews:**
   Collaborate with maintainers to refine and improve your submission.
7. **Stay Engaged:**
   Participate in Mozilla community forums, discussions, and events to stay updated and contribute regularly.

Q5. Explain how data can be persisted in a Docker container using Docker volumes and bind mounts. (BL3, LO4)

1. Docker Volumes:

Managed by Docker and stored in Docker's internal storage area. They are ideal for data that must persist beyond the container lifecycle.

- Example:
- docker volume create myvolume
  docker run -v myvolume:/var/lib/mysql mysql

Volumes are portable, secure, and easily backed up.

- **Avantages:**
  - Managed completely by Docker.
  - Easy to back up or migrate.
  - Data remains intact across container restarts.
  - More secure and isolated from the host system.

2. Bind Mounts:

Maps a directory on the host machine to a directory inside the container. Useful for development and testing.

- Example:
- docker run -v /host/data:/container/data nginx

Changes on the host are immediately reflected in the container.

- **Advantages:**
  - Host and container share the same data.
  - Real-time synchronization between container and host.
  - Useful for development and debugging.